

Algorithmic Scoring & Fit

Probability Matrix

Architectural Overview

The Taledge Fit Score is a deterministic, multi-variate probabilistic calculation synthesized across a dual-track ecosystem. This document outlines the exact mathematical matrices required to calculate localized Success Probability for both **Track 1 (Placement)** and **Track 2 (Competitive Exams)**, ensuring perfect alignment with overarching talent intelligence objectives.

1. Track 1: Placement & Career Success Pathway

In the placement pathway, the platform focuses on preparing students for real-world hiring scenarios. The scoring matrix evaluates professional readiness via 5 critical vectors.

1.1 Vector A: Technical Interview Execution

- Accuracy and Coverage:** Includes Tech Accuracy Score (ratio of correct answers) and Difficulty Weighted Accuracy (harder questions carry exponential weight).
- Problem Solving Depth:** Evaluates Solution Correctness, Approach Structure (decomposition logic), and Multi-Approach Capability (did they propose alternative continuous solutions?).
- Thinking Quality:** Measures Reasoning Clarity, Conceptual Correctness (lenient underlying concept understanding), and Error Recovery.
- Coding Execution:** Assesses Code Correctness, Efficiency (algorithmic complexity), and Readability.
- Micro-Expressions:** Tracks Response Latency, Latency Variance, Hint Dependency, and Consistency over time.

1.2 Vector B: Resume & Profile Features

- Skill Matching Engine:** Cosine similarity between resume skills and JD requirements.
- Project Quality Index:** Semantic analysis of Project Relevance (real-world vs academic), Complexity, and Quantified Impact Outcomes.
- Academic & Quality Signals:** Longitudinal Grades Trend, Education Tier Score, Resume Clarity, and Specificity.

1.3 Vector C: DNLA Social Competence Foundation (Shared Vector)

Integrated via enterprise API endpoints from Germany, this establishes the psychometric baseline.

- **Sub-Matrices:** Achievement Dynamics (Motivation), Interpersonal Relations (Empathy), Will to Succeed (Initiative), and Stress Capacity (Feedback reaction, Resilience).

1.4 Vector D: Behavioural Interview Execution

Explicitly tests the DNLA baseline against conversational realities.

- **Communication Efficacy:** Clarity, STAR approach structure, and Verbosity.
- **Content Quality:** Relevance to intent, Specificity (real examples), and Impact Orientation.
- **Ownership & Attitude:** "I vs Team vs Them" locus of control, and Blame vs Accountability indexes.
- **Consistency Checks:** Alignment with resume claims and internal consistency across responses.

1.5 Vector E: Cross-Component Features

Identifies mathematical disconnects.

- **Tech vs Resume Gap:** High claims + Low performance = Red Flag.
 - **Confidence vs Accuracy Gap:** Overconfidence = Red Flag.
 - **Behaviour vs Psychometric Alignment:** Do verbal actions match claimed DNLA personality traits?
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2. Track 2: Competitive Exam Success Pathway

The competitive exam pathway is fundamentally different in both design and intent. It introduces a multi-layered evaluation model tailored to long-duration, high-stakes preparation environments where standard scoring is insufficient.

2.1 Layer 1: DNLA Success Factor Assessment

Provides the baseline understanding of motivation, discipline, emotional stability, and self-responsibility.

2.2 Layer 2: Counselling-Led Diagnostic Matrix

Unlike Track 1's technical rigor, the AI operates an exploratory diagnostic matrix. It extracts deeper psychological patterns rather than technical

accuracy.

- **Stress & Emotional Mapping:** Identifies baseline anxiety and emotional regulation capabilities under sustained pressure.
- **Preparation Consistency Tracking:** NLP evaluation of study habits, structural discipline, and coping mechanisms during failure cycles.

2.3 Layer 3: Exam Context Mapping

The third computational layer maps the candidate's profile against the specific demands of the chosen examination (e.g., Civil Services vs. GATE).

- Different exams require different forms of endurance and cognitive engagement. The system mathematically aligns the behavioural profile with these exact situational demands.

2.4 Behavioural Risk Indicator Analysis

A critical feature of the Track 2 algorithm is the extraction of **Behavioural Risk Patterns**.

- **Metrics Tracked:** Declining motivation, inconsistent preparation trajectories, difficulty handling feedback, and elevated stress indices.
- **System Action:** These are flagged as non-clinical early awareness indicators. The platform aggregates these trends into specialized dashboards to allow Competitive Exam Institutes to move from reactive intervention to proactive, data-driven support mechanisms.

3. Final Success Probability Calculation

The ultimate output of the platform is customized based on the active tracking pathway.

3.1 Placement Success Probability (Track 1)

Synthesized to predict hiring alignment:

$$\text{Placement Fit Score} = (\text{Tech_Avg} * 0.45) + (\text{Resume_Avg} * 0.15) + (\text{DNLA_Avg} * 0.20) +$$

$$(\text{Behav_Avg} * 0.20) - (\text{RedFlags})$$

3.2 Exam Success Potential Score (Track 2)

Synthesized to predict longitudinal endurance and outcome achievement:

$$\begin{aligned} \text{Exam Potential Score} = & (\text{DNLA_Baseline} * 0.35) + \\ & (\text{Coping_Mechanism_Avg} * 0.35) + (\text{Exam_Context_Match} \\ & * 0.30) - (\text{Risk_Flags}) \end{aligned}$$

By explicitly bifurcating the computational models, the Taledge platform acts as a true Dual-Track Success Intelligence Platform—measuring, predicting, and improving human potential across both careers and competitive pursuits.